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Dotplotic: a lightweight visualization tool for BLAST + alignments and genomic annotations

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Abstract

With the development of sequencing technologies, chromosome-level genome assemblies have become increasingly common across various organisms, including non-model species. BLAST+ is one of the most widely used bioinformatics tools for computing sequence alignments, offering numerous optimizations for speed and scalability. Dot plots, which visualize the similarity between two sequences, are widely used in biological research. However, while many dot plot-generating programs exist, most rely on their own alignment algorithms, and it is uncommon to visualize external BLAST results directly. Here, we present Dotplotic, a lightweight Perl program that generates dot plot-like visualizations based on BLAST output in tabular format. Dotplotic visualizes each alignment as a line connecting the start and end points of the query and subject sequences, with a gradient color indicating sequence identity. It allows users to overlay annotation data from external files onto the plot. Although command-line-based, Dotplotic is implemented as a single script using only core Perl modules, making it easy to install and run across platforms. The program supports standard input for both BLAST results and annotation files, enabling flexible visualization under various conditions, such as filtering specific alignments or displaying selected genomic features like genes or repeats. Dotplotic is an efficient, portable, and easy-to-use visualization tool that enhances the exploration of sequence alignments and serves as a valuable resource for both bioinformatics and broader biological research.

Keywords Dot plot, Perl, BLAST, Genome comparison

Background

Whole-genome sequencing has recently been conducted on a wide range of organisms due to advances in high-throughput technologies such as Oxford Nanopore and PacBio HiFi. Chromosome-level assemblies with high accuracy are becoming increasingly common—not only for the small genomes of viruses and prokaryotes but also for the large, repeat-rich genomes of vertebrates. Comparing assembled sequences, including non-coding regions such as introns and repeats, allows researchers to uncover genome



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diversity and evolution across species. For example, syntenic patterns, such as gene order on chromosomes, are becoming increasingly important in phylogenetic studies e.g., [1].

Dot plots are a widely used method for visualizing sequence comparisons and are helpful to researchers across many biological fields. For bioinformaticians, dot plots between assembled and reference sequences help assess the quality of sequencing and assembly. For evolutionary biologists, they reveal genomic rearrangements: large regions of similarity appear as diagonal lines, repeats generate multiple lines, inverted regions appear as lines in the opposite direction, and insertions or deletions appear as gaps.

In a basic dot plot, dots are placed where the two sequences share the same nucleotide or amino acid on a two-dimensional matrix. However, this simple method is unsuitable for large-scale sequence comparisons. Modern approaches instead calculate similarity between sequences and place dots in regions of high similarity. Several programs follow this approach, such as Dotter [2], JDotter [3], tuple_plot [4], Gepard [5], mummerplot [6], FlexiDot [7], and D-Genies [8]. Because calculating sequence similarity is computationally intensive—particularly for searches involving many long sequences—these tools have improved their algorithms to reduce processing time and memory usage. Cabannes and Klopp [8] classified dot plot-generating tools into two generations based on their user interface: first-generation tools run only on Unix systems and produce static graphics without interactive features, and second-generation tools are platform-independent, user-friendly, and provide interactive interfaces, such as options for changing sequence orientation. The first category includes tuple_plot, mummerplot, and FlexiDot, while Dotter, Gepard, JDotter, D-Genies belong to the second.

Pairwise sequence alignment is a fundamental method for identifying similar regions between two sequences, providing detailed local alignment information. While various tools for sequence similarity searches have been developed, such as minimap2 [9], MMseqs2 [10], and DIAMOND [11], the Basic Local Alignment Search Tool (BLAST) is one of the most commonly used algorithms for sequence similarity searches, despite its long history [12, 13]. BLAST+, a suite of command-line tools developed by the National Center for Biotechnology Information (NCBI), is widely used by both biologists and bioinformaticians [14]. It includes options for parallelization and for reducing database size by masking repetitive regions, making it suitable for large-scale searches on servers. Nearly all DNA sequences generated by researchers are stored in databases maintained by NCBI, and BLAST searches against this vast repository can be run through the NCBI website, contributing greatly to advances in biological and medical research.

Many software tools support BLAST searches and some generate dot plots based on BLAST output, including Unipro UGENE [15], Geneious [16], and NBLAST [17]. However, dot plot visualization tools that directly use external BLAST results remain scarce. Tools like Dotter rely on their own search algorithms and are not optimized to process BLAST output. Graphical user interface (GUI)-based software such as UGENE is user-friendly but not well-suited for large-scale comparisons involving numerous chromosome-level sequences. Dotplotter, a command-line interface (CLI)-based tool that can read external BLAST results [18], offers limited functionality—for instance, it does not display sequence identity in the plots.

Here, we present Dotplotic, a command-line visualization tool that reads external BLAST output and generates dot plots. Dotplotic displays each alignment as a line colored according to sequence identity and allows users to overlay annotation data from

external files. Output is generated in Scalable Vector Graphics (SVG) format, which can be edited with software such as Inkscape and exported to Portable Document Format (PDF). We also developed two supplementary programs: EditDotplotic and Minimapsam2blast6. EditDotplotic allows users to modify SVG files generated by Dotplotic, for example, by rotating query sequence labels. Minimapsam2blast6 converts SAM format files produced by minimap2 into the BLAST tabular-like format files compatible with Dotplotic. Additionally, we provide DotploticGUI.py, a Python-based wrapper that offers a graphical user interface for Dotplotic. This interface enables users to run Dotplotic through an intuitive GUI without needing to type commands. All tools are available on GitHub and are designed to facilitate efficient and user-friendly visualization of large-scale sequence comparisons.

Implementation

Dotplotic, Minimapsam2blast6, and EditDotplotic are implemented in the Perl programming language (version 5.6 or later), and the former two programs require only core modules List::Util, Getopt::Long, and Pod::Text, and the last one requires no modules. DotploticGUI.py is implemented in the Python programming language (Python 3.12 or later), requiring the three modules, tkinter, subprocess, and threading.

Although many advanced Perl modules for bioinformatic analyses, such as BioPerl [19], are available, their installation can sometimes be challenging depending on the computing environments. To ensure ease of use, we developed the three Perl programs requiring only core modules, allowing them to be used immediately after downloading from the GitHub repository without requiring additional dependencies. DotploticGUI.py is the GUI version of Dotplotic, but it must be launched from the CLI, and the current version can only generate the plot file without displaying it on the window, like other tools. Several Python modules for creating GUIs have been developed, and we used them to develop this tool.

Input data

BLAST + programs (blastn, blastx, blastp, tblastn, and tblastx) support various output formats specified via the 'outfmt' option. Dotplotic supports three of these formats: tabular ('-outfmt 6'), tabular with comment lines ('-outfmt 7'), and comma-separated values ('-outfmt 10'). Dotplotic requires that the BLAST search output be formatted appropriately using the 'outfmt' option. Users can include various fields in the BLAST output, such as 'qaccver' as Query accession version, and 'bitscore' as Bit score. The following nine fields are required for Dotplotic processing: 'qaccver', 'saccver', 'pident', 'bitscore', 'length', 'qstart', 'qend', 'sstart', and 'send', and these fields can appear in any order within the input file. Dotplotic also supports overlaying annotation data on the plot, supporting GFF, BED, or OUT from RepeatMasker formats (www.repeatmasker.org) [20].

Drawing process

Using four values—query start ('qstart'), query end ('qend'), subject start ('sstart'), and subject end ('send')—the program draws a line connecting the points ('sstart', 'qstart') and ('send', 'qend'). The percentage of identical matches ('pident') is converted into a hexadecimal (hex) color code. Three identity thresholds—maximum (Max), middle, and minimum (Min)—are defined along with their corresponding color codes. By default, these

are set to 100% (red, '#FF0000'), 80% (orange, '#FFFF00'), and 60% (lime, '#00FF00'). Identity values between Min and Max are linearly mapped to intermediate colors. Values below Min or above Max are assigned the Min and Max colors, respectively. Rectangles representing annotation features are drawn when annotation data is available.

Output format

Dotplotic produces output in SVG format, which can be opened in web browsers, such as Google Chrome and Mozilla Firefox. If the 'click' option is enabled, the SVG file embeds object information that can be viewed by clicking on graphical elements.

Vector graphic software such as Adobe Illustrator and Inkscape can open SVG files and export them to other formats. When converted to PDF, alignment lines and annotation rectangles remain sharp even when zoomed in. Although Dotplotic offers layout flexibility, the display may become cluttered if the BLAST input file includes too many sequences. To address this, the accompanying Perl program EditDotplotic can be used to adjust visual elements such as font size or remove sequence length labels.

Results

Dotplotic represents each alignment in the BLAST tabular format as a single line, each annotation data as a rectangle (Fig. 1). The Dotplotic workflow consists of four steps (Fig. 2): (1) Given a BLAST output file, and optionally an annotation file, Dotplotic reads the input. (2) Dotplotic calculates basic statistics, including the number and total lengths of query and subject sequences. If the sequence order is not specified by the user, Dotplotic automatically arranges the order so that the general direction of the BLAST alignment appears as a diagonal line from top left to bottom right of the plot. To achieve

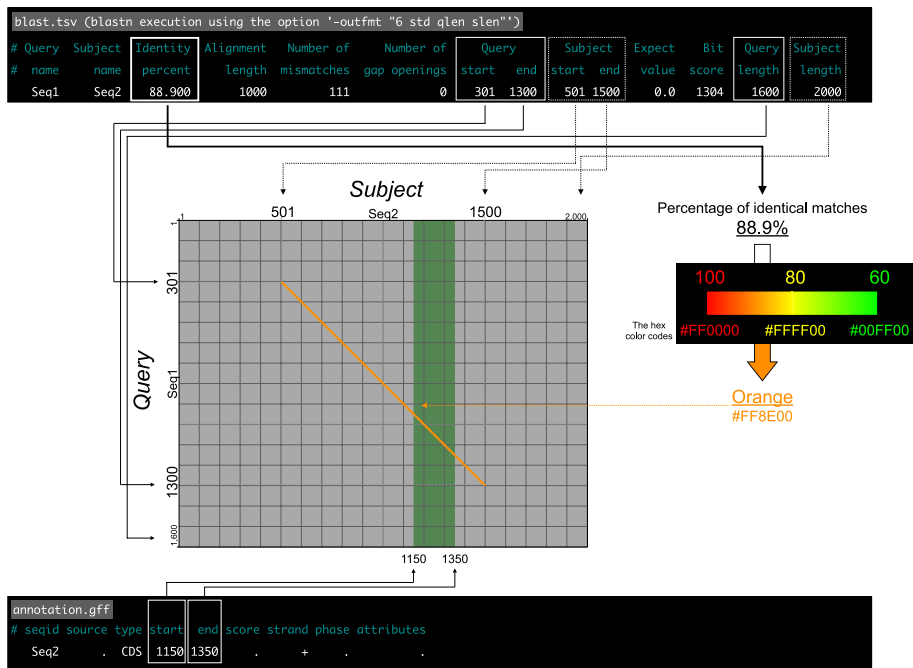


Fig. 1 Workflow for generating plots based on a BLAST alignment in tabular format and annotation data in GFF format. For each BLAST alignment, a line is drawn connecting the start and end positions of the query and subject sequences, with a hexadecimal color code representing sequence identity. A rectangle is also drawn to represent features from the annotation data

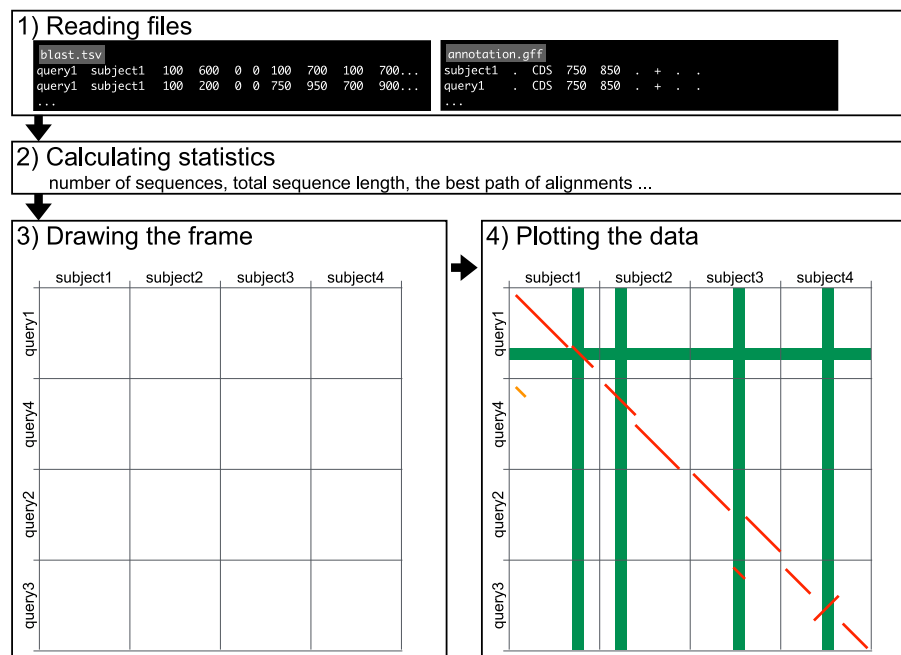


Fig. 2 Overview of Dotplotic. After reading the BLAST file (1), Dotplotic calculates the total sequence lengths and the number of query and subject sequences as well as similarities among them using a best-path search to determine the sequence order (2). Based on these results, frames are drawn (3), followed by the alignment lines and the annotation rectangles (4)

this, best-path searches are performed across all query-subject combinations to determine the query-subject relationships and their relative orientation. Subjects are sorted by name, and queries are ordered according to the order of their associated subjects. For large BLAST files (e.g., genome-to-genome comparisons), best-path computation may consume significant memory and time. In such cases, users can reduce resource usage by manually specifying sequence order or by enabling the ‘light’ mode. When this option is enabled, query and subject sequences are automatically sorted by name, and short BLAST alignments (less than 1kbp) are excluded, thereby reducing both computational load and the size of output SVG file. (3) Based on the statistics of the BLAST alignments, Dotplotic determines the layout of the plot. The reduction scale is set according to the total length of the query or subject sequences, the sequences are arranged in a defined order, and individual plotting regions are allocated for each sequence. (4) If annotation data are provided, they are displayed as rectangles. Finally, BLAST alignments are drawn as colored lines on the plot.

SAM-format alignment data includes the names of the query and subject sequences, the alignment starting position on the subject, and a CIGAR string. The query start and end positions, query length, and the numbers of mismatches, insertions, and deletions can be calculated from the CIGAR string. The dynamic programming (DP) alignment score that is denoted by the ‘AS:i:’ tag in the SAM file is used as the bit score. Although the resulting output reports all E-values as ‘0.0’ because a single SAM alignment does not provide sufficient information to compute them, Minimapsam2blast6 generates alignment results in a BLAST-like tabular format based on this information.

Sequence identity is then calculated based on the derived alignment statistics. The other accompanying program, EditDotplotic, modifies specific objects in the SVG file

generated by Dotplotic. Most of the SVG objects in the output file are represented by individual lines of text, each including an SVG tag and an identifying comment tag. For example, query sequence labels in the plot are written as lines beginning with '<text' and containing the tag '<!--query_name-->'. EditDotplotic uses these comment tags to locate and modify targeted elements within the SVG file. DotploticGUI.py is a supplementary Python program that provide a GUI for executing Dotplotic (Fig. S1). While the GUI is launched from the CLI, it allows users to specify all parameters and paths of input BLAST file and output SVG file without directly editing the command-line options. Although the generated plot is not displayed within the window, the interface facilitates the full configuration and execution of Dotplotic, making it more accessible to users unfamiliar with the CLI.

Discussion

In this study, we present Dotplotic and other accompanying programs. Our aim was to develop a lightweight and easily installable tool for visualizing alignment results with annotation data, without requiring complex dependencies. Minimapsam2blast6 allows to convert minimap2 outputs into a format compatible with Dotplotic. Users can further customize plot features, such as sequence label tilt and alignment line thicknesses, using EditDotplotic. For users unfamiliar with the CLI, the GUI wrapper, DotploticGUI.py, provides an accessible interface. Sequence alignment of long and numerous sequences is often performed on high-performance servers that lack graphical desktop environments, which presents challenges for visualization. Dotplotic, Minimapsam2blast6, and EditDotplotic are compatible with both laptops and headless server environments, providing a practical solution for genome comparison visualization.

Case study in yeast genomes

Here, we demonstrate the visualization of BLAST alignments involving three yeast genomes using Dotplotic. The lager-brewing yeast *Saccharomyces pastorianus* is an interspecific hybrid of *S. cerevisiae* and *S. eubayanus*, and the genome of one of its strains, CBS 1483, has been sequenced at the chromosomal level [21]. We obtained nuclear genome sequence datasets for the three yeast species from GenBank: 31 chromosomes (total length: ca. 23.0 Mbp) for CBS 1483 (*S. pastorianus*, GenBank assembly accession: GCA_011022315.1), 16 chromosomes (total length: 11.6 Mbp) for *S. cerevisiae* (GCF_000146045.2), and 16 chromosomes (total length: 12.1 Mbp) for *S. eubayanus* (GCF_001298625.1). Additionally, we downloaded the genome annotation file of *S. cerevisiae* (GCF_000146045.2_R64_genomic.gff) from GenBank.

A BLAST search was conducted using CBS 1483 as the query and the other two species as subjects, with output in tabular format including query and subject lengths (using '-outfmt "6 std qlen slen"'). Alignments shorter than 1 kbp were filtered out to reduce the output file size (23 MB to 1 MB). Dotplotic was subsequently executed to visualize the BLAST alignments (Fig. 3). All commands used for the BLAST search and for running Dotplotic to generate Figs. 3, 4, and 5 are listed in Supplementary Table 1. EditDotplotic was also used to modify the sequence names in the Dotplotic outputs.

Genomic rearrangements have been reported in several chromosomes [21], and we visualized these events using Dotplotic (Fig. 4). Homologous recombination between chromosome X of *S. cerevisiae* and *S. eubayanus* was clearly observed in CBS 1483

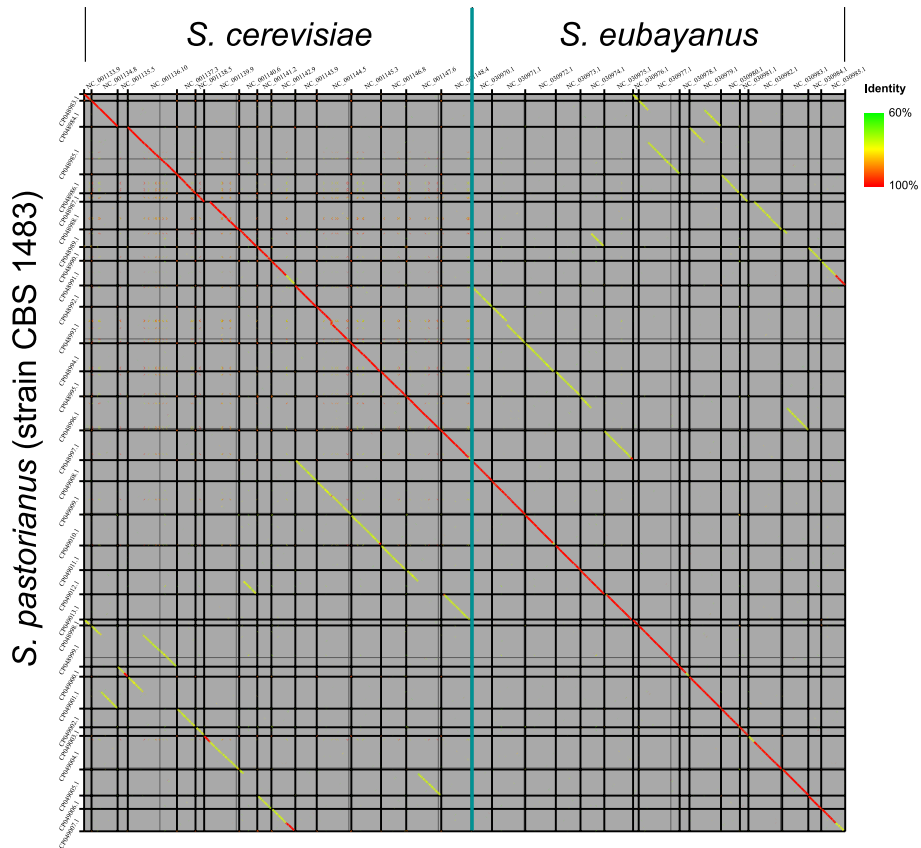


Fig. 3 The genome comparison of yeast genomes using Dotplotic

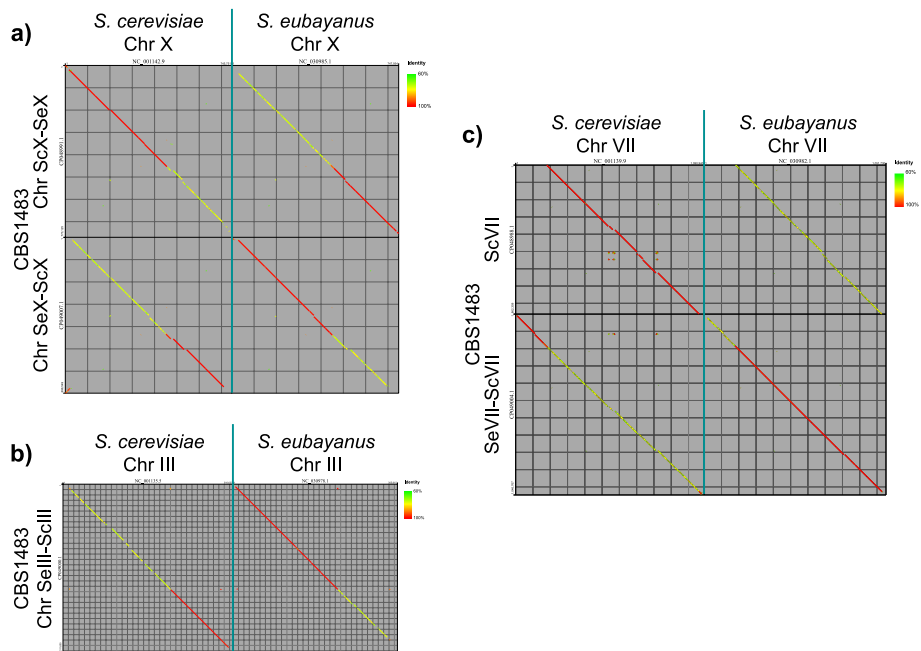


Fig. 4 Comparisons of yeast chromosomes using Dotplotic. Chromosomes are abbreviated as "Chr"

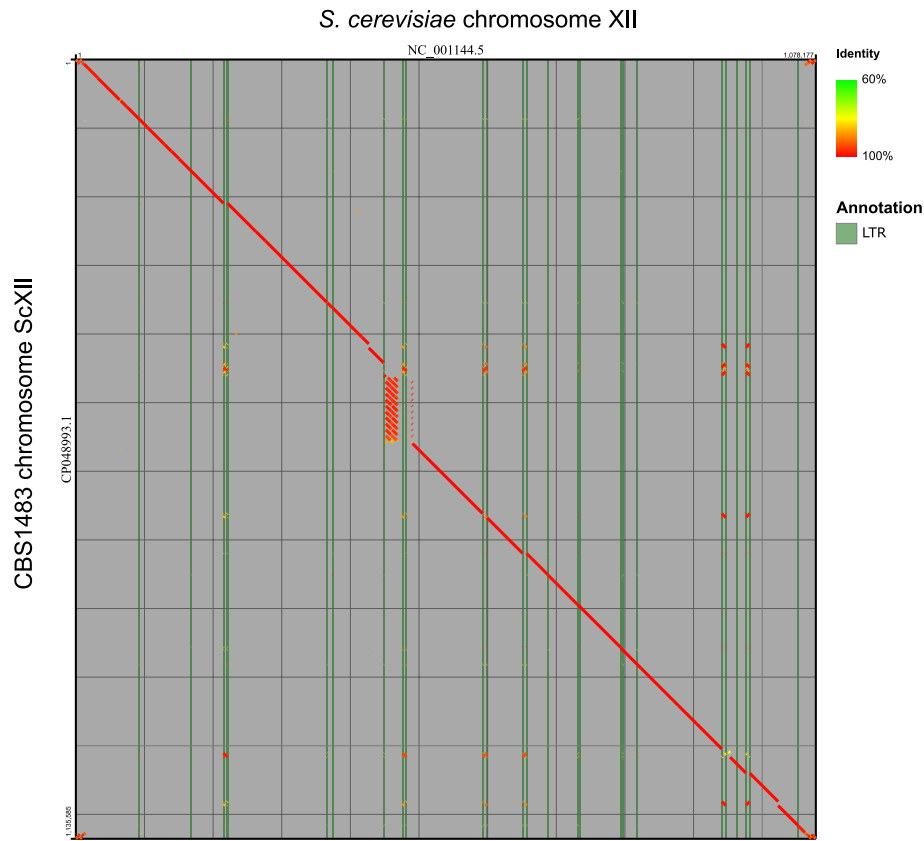


Fig. 5 Comparisons of yeast chromosomes using Dotplotic. Chromosome ScXII from *Saccharomyces pastorianus* strain CBS1483 was compared with chromosome XII from *S. cerevisiae*. Long terminal repeats (LTRs) in *S. cerevisiae* chromosome XII are shown as rectangles

(Fig. 4a). In chromosome SeIII-ScIII of CBS 1483, approximately two-thirds of the first half showed high similarity to chromosome III of *S. eubayanus*, while about one-third of the second half matched chromosome III of *S. cerevisiae*, suggesting a homologous recombination event between the two species (Fig. 4b). However, the truncated regions, approximately two-thirds of the first half of chromosome III in *S. cerevisiae* and about one-third of the second half of chromosome III in *S. eubayanus*, were not present in the CBS 1483 assembly (Fig. 3). While chromosomes ScVII and SeVII-ScVII in CBS 1483 correspond to chromosome VII in *S. cerevisiae* and *S. eubayanus*, respectively, approximately 100 kbp from the first half of *S. cerevisiae* chromosome VII align with that of SeVII-ScVII in CBS 1483, suggesting a translocation from *S. cerevisiae* to *S. eubayanus* (Fig. 4c). Although chromosome ScXII in CBS 1483 was generally similar to chromosome XII in *S. cerevisiae*, several repeated regions were observed (Fig. 5). The repetitive alignments in the central region suggest tandem repeat insertions. Additionally, all interspersed alignments were flanked by long terminal repeats (LTRs) on both the 5' and 3' ends, suggesting the proliferation of these regions along with LTR expansion.

Case study in other species (*Drosophila*, *Arabidopsis*, and *Escherichia coli*)

We also demonstrate the application of Dotplotic for genome comparison other species, including 2 *Drosophila* species (*D. melanogaster* and *D. simulans*), 2 *Arabidopsis* species (*A. thaliana* and *A. arenosa*), and 2 *Escherichia coli* strains (K-12 and O157:H7)

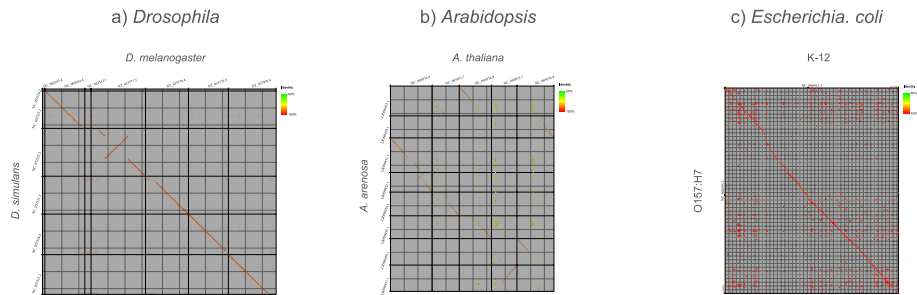


Fig. 6 Genome comparisons of various species, generated using Dotplotic. Short alignments (less than 2kbp) were excluded from all plots. EditDotplotic was used to rotate sequence labels in the *Drosophila* and *Arabidopsis* comparisons, and to adjust the alignment stroke width in *Arabidopsis* comparison

Table 1 Comparison of dot plot-generating tools						
Tool	Interface type	Built-in alignment program	Multi-to-multi comparison	Interactivity on dotplot	Overlay information	Citation
Gepard	GUI (CLI available)	Suffix array calculation	No Labels	Zoom, filtering	No	Krumsiek et al. [8]
Unipro UGENE	Native GUI	Suffix array calculation	No Labels	Zoom	No	Okonechnikov et al. [12]
MUMmer4	Command line only	nucmer, promer	Labels	Static image	No	Marçais et al. [10]
D-GENIES	Web-based GUI (launched via CLI)	Minimap2	Labels	Zoom, filtering	No	Cabanettes and Klopp [4]
Dotplotic	CLI (with optional Python GUI)	None	Labels	Optional data display	Available	This study

(Fig. 6). As with the yeast genome comparisons, BLAST searches were conducted using only nuclear genomes. All command lines used to run Dotplotic, the number of BLAST alignments, computing performance, and output file sizes are summarized in Table S2. The output images with the smallest file size for each species were converted to PDF format and shown in Fig. 6.

Running Dotplotic on large BLAST alignment datasets increases memory usage, processing time, and the size of resulting SVG output file. Although converting SVG files into PDF format using Inkscape remarkably reduced the file size, the resulting PDFs still exceed 1 MB when the alignment count approaches one hundred thousand. Dotplotic’s best path search running for calculation of relative distances between query and subject sequences is skipped when the ‘light’ option is enabled. This substantially reduces both memory usage and processing time. When a BLAST result contains a large number of short, potentially uninformative alignments, filtering them can be an effective approach for generating a more interpretable and concise overview of genome comparisons.

Comparison to other dot plot generating tools

We compared Dotplotic with several recently developed dot plot-generating tools that are available across various platforms, including Gepard [5], Unipro UGENE [15], MUMmer4 [6], and D-GENIES [8] (Table 1). Among these tools, only MUMmer4 is command-line based, while the other three tools, Gepard, Unipro UGENE, and D-GENIES, are primarily GUI-based. These GUI-oriented tools generate dynamic

plots that users can interact with, such as by zooming, offering a high level of interactivity, while the command-line based tool generated static plots. Although users can run Dotplotic through the GUI version (DotploticGUI.py) and access object information by enabling the 'click' option, the output graphic itself is not interactive, suggesting that Dotplotic belongs to the first-generation category defined by Cabanettes and Klopp [8]. However, Dotplotic does not perform alignment searches itself and dedicated solely to visualization. Based on this characteristic, it may be considered as belonging to a third category: visualization-specific dot plot generating tool.

Despite these limitations, we believe that Dotplotic offers three key advantages. First, it supports output from BLAST+, a widely used command-line alignment tool [14]. According to Google Scholar, BLAST+ was cited 2,890 times in 2024 alone, indicating that many bioinformaticians have installed and used the tool. It is likely employed in routine analyses without citation, suggesting that BLAST+ is widely and regularly used, even for minor or background tasks. By providing an intuitive visualization option for BLAST+ results, Dotplotic can significantly contribute to bioinformatics research. Minimap2 is also widely used alignment tool [9]. The accompanying program, Minimap2blast6, convert its SAM-format output into a BLAST-like tabular format, enabling Dotplotic to generate dot plots from the results.

Second, Dotplotic can easily overlay annotation data onto plots. The relationship between LTRs and repetitive regions is visualized in Fig. 5, which was generated with a single command (Table S1), demonstrating its ease of use. By modifying the annotation input, Dotplotic users can investigate the types and locations of genomic rearrangements. It is one of the most beneficial features of Dotplotic is its ability to combine alignment results and annotation data in a single command, without requiring intermediate files or preprocessing by other tools.

Finally, Dotplotic is compatible with a wide range of platforms. Software with advanced features, such as high-throughput computing or multifunctional capabilities, is sometimes sensitive to changes in the development environment and may fail after updates to either the software or the underlying platform. In contrast, Dotplotic is a lightweight Perl program that depends only on core modules, making it likely to remain functional across diverse systems into the future.

Conclusion

BLAST+ is a command-line-based alignment search tool and one of the most widely used programs in bioinformatics. However, most existing dot plot visualization tools rely on their own alignment algorithms, and visualizing results directly from BLAST+ output remains difficult. To fill this gap, we present Dotplotic, a visualization tool specifically designed for BLAST+ output. Dotplotic is implemented as a single Perl script that depends only on core modules, making it lightweight, platform-independent, and easy to use.

Dotplotic can not only generate plots for genome comparisons but also overlay annotation data onto the plots, allowing users to visually compare genomic rearrangements with annotated features. This makes it a valuable tool for many bioinformaticians and biologists.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12859-025-06255-5>.

Additional file 1.

Additional file 2.

Author contributions

Author Contributions Statement H. M. developed the program, and prepared figures. H.M and T.O. wrote the main manuscript text. All authors reviewed the manuscript.

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Availability of data and materials

The sequence data used in the case studies section are available from GenBank: *Saccharomyces cerevisiae* strain S288C (GenBank assembly accession: GCF_000146045.2). *S. pastorianus* strain CBS 1483 (GCA_011022315.1). *S. eubayanus* strain FM1318 (GCF_001298625.1). *Drosophila melanogaster* (GCF_000001215.4). *D. simulans* (GCF_016746395.2). *Arabidopsis thaliana* (GCF_000001735.4). *A. arenosa* (GCA_905216605.1). *Escherichia coli* str. K-12 substr. MG1655 (GCF_000005845.2). *E. coli* O157:H7 str. Sakai (GCF_000008865.2). The annotation data used in the case study for *S. cerevisiae* is available from the NCBI FTP site: *S. cerevisiae* strain S288C (the file name: GCF_000146045.2_R64_genomic.gff) (URL: https://ftp.ncbi.nlm.nih.gov/genomes/all/GCF/000/146/045/GCF_000146045.2_R64/).

Declarations

Competing interests

The authors declare no competing interests.

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References

- Schultz DT, Haddock SHD, Bredeson JV, Green RE, Simakov O, Rokhsar DS. Ancient gene linkages support ctenophores as sister to other animals. *Nature*. 2023. <https://doi.org/10.1038/s41586-023-05936-6>.
- Sonnhammer ELL, Durbin R. A dot-matrix program with dynamic threshold control suited for genomic DNA and protein sequence analysis. *Gene*. 1995;167(1–2):GC1–10.
- Brodie R, Roper RL, Upton C. JDotter: a Java interface to multiple dotplots generated by dotter. *Bioinformatics*. 2004;20(2):279–81.
- Szafranski K, Jahn N, Platzer M. tuple_plot: fast pairwise nucleotide sequence comparison with noise suppression. *Bioinformatics*. 2006;22(15):1917–8.
- Krumsiek J, Arnold R, Rattei T. Gepard: a rapid and sensitive tool for creating dotplots on genome scale. *Bioinformatics*. 2007;23(8):1026–8.
- Marçais G, Delcher AL, Phillippy AM, Coston R, Salzberg SL, Zimin A. MUMmer4: A fast and versatile genome alignment system. In: Darling AE, editor. *PLoS Comput Biol*. 2018;14(1):1005944. <https://doi.org/10.1371/journal.pcbi.1005944>.
- Seibt KM, Schmidt T, Heitkam T. FlexiDot: highly customizable, ambiguity-aware dotplots for visual sequence analyses. In: Hancock J, editor. *Bioinformatics*. 2018;34(20):3575–7.
- Cabanettes F, Klopp C. D-GENIES: dot plot large genomes in an interactive, efficient and simple way. *PeerJ*. 2018;4(6):e4958.
- Li H. Minimap2: pairwise alignment for nucleotide sequences. *Bioinformatics*. 2018;34(18):3094–100.
- Steinegger M, Söding J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nat Biotechnol*. 2017;35(11):1026–8.
- Buchfink B, Reuter K, Drost HG. Sensitive protein alignments at tree-of-life scale using DIAMOND. *Nat Methods*. 2021;18(4):366–8.
- Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ. Basic local alignment search tool. *J Mol Biol*. 1990;215(3):403–10.
- Altschul S. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res*. 1997;25(17):3389–402.
- Camacho C, Coulouris G, Avagyan V, Ma N, Papadopoulos J, Bealer K, et al. BLAST+: architecture and applications. *BMC Bioinformatics*. 2009;10(1): 421.
- Okonechnikov K, Golosova O, Fursov M, the UGENE team. Unipro UGENE: a unified bioinformatics toolkit. In: *Bioinformatics*. 2012;28(8):1166–7.
- Kearse M, Moir R, Wilson A, Stones-Havas S, Cheung M, Sturrock S, et al. Geneious Basic: an integrated and extendable desktop software platform for the organization and analysis of sequence data. *Bioinformatics*. 2012;28(12):1647–9.
- Choi BS, Choi SK, Kim NS, Choi IY. NBLAST: a graphical user interface-based two-way BLAST software with a dot plot viewer. *Genomics Inform*. 2022;20(3): e36.
- Mohite OS, Jørgensen TS, Booth TJ, Charusanti P, Phaneuf PV, Weber T, et al. Pangenome mining of the *Streptomyces* genus redefines species' biosynthetic potential. *Genome Biol*. 2025;26(1):9.
- Stajich JE, Block D, Boulez K, Brenner SE, Chervitz SA, Dagdigan C, et al. The Bioperl toolkit: Perl modules for the life sciences. *Genome Res*. 2002;12(10):1611–81.

20. Smit A, Hubley R, Green P. RepeatMasker Open-4.0. 2013; Available from: <http://www.repeatmasker.org>.
21. Salazar AN, Gorter De Vries AR, Van Den Broek M, Brouwers N, De La Torre CP, Kuijpers NGA, et al. Chromosome level assembly and comparative genome analysis confirm lager-brewing yeasts originated from a single hybridization. *BMC Genomics*. 2019;20(1):916.
22. Shen W, Le S, Li Y, Hu F. 2016 SeqKit: A cross-platform and Ultrafast toolkit for FASTA/Q file manipulation. In: Zou Q, editor. *PLoS ONE*. 11(10):e0163962.

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